

1 · bank preparation (deterministic, once)
AudioMNIST 48 kHz → resample 16 kHz · peak-norm 0.5
1% energy trim · true lengths kept

2 · noise (per clip draw)
 $x \leftarrow x + \eta, \eta \sim N(0, \sigma^2), \sigma = \text{RMS}(\text{speech}) \cdot 10^{(\text{dB}/20)}$
(0 dB primary condition, +5 dB robustness companion)

3 · frontend (one of three pathways → Fig. G8)
envelope: STFT→mel→log→affine (62.5 fps) · quadrature: + baseband phase
carrier: FFT band-split 96–1,536 Hz (16 kHz) → rows $[T \times 16]$

4 · gain (calibrated per frontend)
 $\text{rows} \leftarrow g \cdot \text{rows}$
validity: $\max_{\text{row}} |\text{dt} \cdot g \cdot \text{row}| < \pi$ · kick stats recorded per cell

5 · tonotopic broadcast
band $r \rightarrow$ every oscillator of grid row r , all C channels
(per-shape tonotopic mapping pre-declared)

6 · field integration (explicit Euler, $\text{dt} = 0.1, T$ steps)
 $\theta_{\text{row}} = \omega_{\text{row}} + \text{coupling}_{\text{row}}(\theta; K) + \text{drive}_{\text{row}} - \lambda \cdot \sin \theta_{\text{row}}$
coupling evaluated spectrally: $K_{\text{row}} = \text{FFT}(K)$ under the boundary geometry's wrap
(torus / cylinder / sheet / helix / cube / sphere — the shape factor lives here)
spectral clamp caps $|K_{\text{row}}|$ · coupling law per physics family · SL adds amplitude ODE

7 · state trajectories
 $(\sin \theta(t), \cos \theta(t))$ for all 1,024 oscillators · T frames
valid-length mask carried throughout

8a · readout (the verdict path)
masked pooled statistics → $\Phi(x)$:
full-span · 4-window · macro · parity · settle
ridge probe $W = (\Phi_{\text{row}} \Phi + \lambda I)_{\text{row}}^{-1} \Phi_{\text{row}} Y$
floor control: same ridge, field bypassed
(step 3 output → step 8a directly)
accuracy → pre-registered bars

8b · instruments (mechanism path)
drive-phase PLV: $|\text{row} \cdot e^{i(\theta - \phi)}|$ over time,
 ϕ = Hilbert phase of the delivered drive
order parameter $R = |\text{row} \cdot e^{i(\theta)}|$ over oscillators
entrained fraction (PLV > 0.5)
kick stats (bound verification)
diagnostics — never verdicts

every cell records command + commit + seed · accuracies · instruments · kick stats;
bars pre-registered before launch, scored only at protocol n (2,048 / 512 clips)